

SECTION A : DATA Fundamentals

1. A Database is essentially a place where data is stored, retrieved and manipulated. It is comprised of a structure inside, the database, schema, table. Examples of where databases can be found are home affairs and insurance companies (Discovery)
2. A Datawarehouse is a central place where data that has been through the ETL process is stored. It is data that has been cleaned coming from multiple sources. Databases use an online transactional Processing system that stores data in real time. Data warehouses use an online analytical process system.
3. Data Lake is essentially a large storage space where business store all their raw and unprocessed data. It usually comes in multiple different formats and businesses usually decide later what to do with it. It is different from a datawarehouse because a warehouse stores clean data and it is organised and ready for analysis. A business may prefer a data lake when it has large amounts of data, it comes in different formats, they are not yet sure what they would need or use it for, future analysis, and machine learning.

4. A Data mart is a small part within a datawarehouse that houses specific departmental data. For example Finance department, sales department. It is one small section inside the datawarehouse. It's good because departments can focus on data they need instead of the large volume from the entire warehouse.

5. Structured: usually data that comes in tabular format. It is comprised of columns and rows. Data in excel

Semi-structured: Data that doesn't follow a strict structure, it has labels. JSON

unstructured: Data that is free-form, doesn't follow a specific structure. It can come in all forms, social media, PDF.

6. Data modelling is a way of organising the data and showing how it connects (all moving parts)

Examples include star schema: This is a model where the fact table remains at the centre and is surrounded by dimension tables. Snowflake schema, dimensions are further broken down into more related tables

7. Data mining is essentially looking at a large amount of data in order to discover patterns, trends and useful insights. Businesses use it to uncover hidden trends or relationships to potentially capitalize. Examples include, Netflix, to understand viewer behaviour, Airlines, analyse booking trends and patterns to adjust pricing.

8. A DBMS is a software that allows users to interact with databases in order to store, retrieve, clean and interact with data. Examples include: SQL Server, Databricks, snowflake, mysql

9. ETL is essentially the process of cleaning, extracting and loading data through a pipeline into a data warehouse for analysts. Extract: The data is extracted from multiple data sources in its raw format. Transform: The process of cleaning, preparing the data. Removing duplicates. Loading: sending the data to the warehouse after being cleaned.

10. CSV: Data stored in rows and column format, separated by commas

JSON: Data that has labels & keys

Parquet: stores large amount of data

Avro: stores data and explains what the data type should be

11. AI: A software that can perform activities and tasks that would normally require human intelligence. Things such as learning, finding patterns, understanding language, Decision making.

Narrow AI: It is the AI that is good at one specific thing. example includes face recognition software. only does one job

General AI: AI that can learn and perform a wide range of intellectual tasks like a human. example, telling Claude that I want to plan a business.

12. Machine Learning is a ^{branch} type of AI where computers learn patterns from data and use those patterns to make predictions or decisions. Traditional Programming is a method where a Programmer explicitly writes rules and instructions that a Computer follows to produce an output.

Supervised Learning: A type of ML where a model learns from labelled data, using known answers to make predictions on new data.

Unsupervised Learning: A different type that uncovers trends, patterns and relationships without labelled answers.

Reinforcement Learning: A 3rd type where the agent follows certain line of action and receives a reward or penalty based on those actions.

13. Deep Learning: A machine learning that uses multi-layered neural networks to learn complex patterns from large amounts of data. It has input layers, hidden layers, output layers. Input: receives the data
hidden: Process the data and find patterns
output: Produces the answer.

14. A branch of AI that enables computers to understand, interpret and generate human language. Voice assistants: Siri, Alexa, Google Assistant
Translation: Google Translate uses NLP to translate text from one language to another
Spam detection: Email services use NLP to identify and filter unwanted or spam messages.

15. LLM (An AI model trained on huge amounts of text so it can understand and generate human-like language.

Examples: ChatGPT - used for answering questions, writing, summarising, coding and translate

Google Gemini: used for answering questions, generating text, analysing information, and assisting with tasks

16. Generative AI: A type of AI that can create new content such as text, images, music, video or code.

At a higher level, it is trained on large amounts of existing data. It learns patterns and relationships in that data

Examples: Github Co-Pilot
ChatGPT

17. A training dataset is a collection of data used to teach an AI model how to recognise patterns and make predictions or decisions. The AI learns directly from the training data. If the data is accurate, relevant and representative, the model is more likely to produce accurate results.

18. Computer vision: A branch of AI that allows computers to understand and interpret images and videos.

Machines don't see like humans. They process an image as numerical data. AI models analyse these pixels to identify patterns, shapes, objects, faces or other features and then determine what is in the image.

19. An API is a way for two different software applications to communicate with each other and exchange information or services. A real world analogy would be, you build a weather app, the app doesn't have its own weather sensors or weather database, instead it sends a request to a weather service through its API

20. A rest API is an API that allows different software systems to communicate over the internet using standard HTTP requests. REST stands for Representational state Transfer.

GET : retrieves data

POST : Creates new data

PUT : updates existing data

DELETE : removes data

22. JSON is a simple format for storing and exchanging data between computer systems. JSON is commonly used because it is easy for humans to read, easy for computers to process. It is lightweight meaning it doesn't store unnecessary information.

23. Prompt Engineering is the process of creating clear and specific instructions to get better and more useful results from an AI tool. A vague prompt can produce a vague or unhelpful answer, while a clear prompt gives the AI more information about exactly what you want. Examples, Poor Prompt: Explain databases. Improved Prompt: Explain the difference between SQL and NoSQL databases in simple terms.

24. A map function takes each item in a collection, applies a function to it, and produces a new collection containing the results. In programming and data processing, it is useful when you want to perform the same transformation on every piece of data.

25. Cloud Computing is the delivery of computing resources over the internet instead of having to own and manage the physical hardware yourself. The three main cloud services include:

- * IaaS - Infrastructure as a service: Provides basic

computing infrastructure such as virtual machines, storage, and networks. You manage the operating system and applications. I.e. Amazon EC2

- * PaaS - Platform as a service: Provides a platform where developers can build and run applications without managing the underlying servers and infrastructure. Example: Google App Engine.

- * SaaS: software as a service: Provides ready to use software over the internet. You simply use the application, the provider manages everything behind it. Example: Microsoft 365

26. Version Control systems (VCS): A version control system is a tool that tracks changes made to files over time. It allows you to see what changes were made, go back to an earlier version, work with other people without overwriting each other's work

27. Jupyter Notebook is an interactive environment where you can write and run code, view the results, and add text, and explanations in the same document. Most commonly used language is Python. It is popular because it allows them to: write and run code step-by-step, analyse and clean data, create charts and visualisations, see results immediately document findings along their code.

28. Python is a high-level programming language used to create software, automate tasks, analyse data and build AI systems. It is popular because it has simple, readable syntax, it has many libraries specifically designed for data analysis and AI, it can handle large amounts of data, it is widely supported and has a large community. Four common Python libraries include: Pandas, NumPy, Matplotlib, SciKit-learn.

29. Data Privacy is the protection of a person's personal information and their right to control how that information is collected, stored, used and shared. Examples of personal data include name, email, ID number, location and IP Address. It's important because it protects people from misuse of their personal information.

30. Data ethics is the set of principles that guide how data should be collected, used, stored, and shared in a fair, responsible, and respectful way. Examples:

1. Collecting data without permission
2. Using biased AI for hiring
3. Selling people's personal information without people's knowledge.

31. Data governance is the system of rules, policies, roles, and processes an organisation uses to manage its data properly. Its purpose is to make sure data is accurate, secure, consistent, accessible and used responsibly.

Key components:

- Data Policies - Rules for how data should be collected, stored, accessed.
- Data quality - Ensures data is accurate, complete and consistent
- Data security - Protects data from unauthorised access, loss or theft
- Roles & responsibilities - Defines who is responsible for managing & protecting
- Data Privacy - Ensures personal and sensitive information is handled properly
- Data standards - Establishes consistent formats and definitions so data is understood and used consistently.

32. Bias in AI occurs when an AI model consistently produces unfair or inaccurate results for certain people or groups. An AI model becomes biased when the data it was trained on, the way it was designed, or the way it is used causes it to favour or disadvantage certain groups. Real world example would be AI hiring system, it is trained using historical hiring data from a company where most successful applicants were men. The AI learns this pattern and starts giving male applicants higher scores than equally qualified female applicants.

33.

~~Data files~~

33.

Data Roles

Role	Main Role	Key skills	Tools
Data Analyst	Analyses existing data to find trends and provide insights for business decisions	SQL, Excel, data visualisation	Excel, SQL, Power BI, Tableau
Data scientist	uses data, statistics, and Machine learning to make predictions and solve complex problems	Python, statistics, Machine learning, SQL	Python, Pandas, Jupyter, scikit-learn
Data engineer	Builds and maintains the systems & pipelines that collect, transform, and store data	SQL, Python, databases, ETL, data pipelines	Python, SQL, spark, Airflow, cloud platforms

34. Business intelligence is the process of using data to understand what is happening in a business and make better decisions. BI tools take raw business data, analyse it, and turn it into useful reports, dashboards, and visualisations.

Two Popular BI Tools:

- Power BI : Helps businesses identify trends, track performance
- Tableau : Helps businesses see patterns and trends that may be difficult to notice in raw data

35. How the world of Data is broad and consists of many different moving parts, not just a single line of things and how they're all connected.

36. Machine Learning things are very complicated.
I watched Youtube videos

37. Data and AI will have a major impact on my future career and everyday life. In my career, I will use data to analyse business performance, identify trends, and help companies make better decisions. As a data Analyst, I could use Power BI and SQL to analyse sales data and identify which products are performing well.